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# A Benchmarking Study of Machine Learning and Deep Learning Models for Long-Term Time Series Forecasting

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## Group 6

### Abstract

Long-term time series forecasting is widely used in applications such as electricity analysis and weather prediction, but fair comparison across forecasting models remains challenging. This paper presents a reproducibility-oriented benchmark of four representative forecasting paradigms, including dedicated traditional (DLinear) and machine-learning-oriented (LightGBM) methods, as well as two deep-learning-based models, LSTM and PatchTST. On ETTm1, Electricity, and Weather, each model was evaluated under four prediction horizons ( $H = 96, 192, 336, 720$ ). All experiments were either reproduced or adapted from Time-Series-Library using a common reporting format, including MAE, RMSE, MSE, MAPE, prediction horizon, dataset, and sample split statistics. The results show that: LightGBM is the standout performer overall numerically (and well ahead of the other strong competing models), PatchTST is the strongest deep learning model and LSTM provides an extremely simple recurrent neural baseline while DLinear here effectively serves simply as trend-oriented comparison. The benchmark demonstrates the advantage of cross-paradigm competition against within-architecture family comparisons for forecasting models.

## 1 Introduction

Long-range time series forecasting is an essential task in electricity analysis, weather prediction, and data-driven planning. Time series differ from other data types, such as images or text, because their informative content is mostly expressed through temporal change (i.e. trend, seasonality, volatility and cross-variable dependencies). As such, forecasting quality depends not only on the architecture of the models used but also on how they behave under different prediction horizons and data conditions.

However, even with the various methods proposed in literature, performing fair comparison proves to be difficult as results that are reported are performed on different datasets, horizons and implementation decisions. The problem is that, although there are many good models and datasets available, their reproducible evaluation across models, datasets, and forecasting tasks is limited by a lack of a common benchmark framework; public benchmark frameworks such as Time-Series-Library [8] can help here. For example, TimesNet [9] is more of a general time series model but illustrates the common necessity of systematic evaluations on multiple time series tasks.

Here we do not propose a new model, but instead a benchmark paper. We first compare four representative paradigms: DLinear (a linear baseline), LightGBM (feature-based machine learning model), LSTM (recurrent neural network) and PatchTST (a Transformer-based model). On ETTm1, Electricity, and Weather: each model evaluated at ( $H = 96, 192, 336, 720$ ). The aim is to study how well different forecasting paradigms cope within a single long-term forecasting framework.

## 2 Related Work

In the field of time series forecasting, linear baselines, feature-based machine learning models, recurrent neural networks, Transformer-based architectures have all been recently reviewed or compared against one another. LightGBM is an efficient gradient boosting framework but can be adapted into a forecasting procedure using lagged observations as supervised learning features [1, 2]. LSTM is a traditional recurrent model that implements complex time-series dynamics through the use of gated memory mechanisms [5, 6] and has been widely used to solve time series problems such as forecasting tasks [7].

Recent forecasting tasks have had more frequent applications of Transformer-based models. PatchTST patches time series and learns long-range temporal structure via Transformer representations [3]; RevIN is designed to mitigate distribution shift of this class of inputs to the task at hand [4]. Moreover, Time-Series-Library gives a consolidated standard on replicable time series benchmark [8] and based on perception of general deep time series models TimesNet were generated to present these works [9]. Based on these studies, this paper conducts the same forecasting setting to compare DLinear, LightGBM, LSTM and PatchTST.

## 3 Benchmark Scope

It focuses on long-term multivariate forecasting. We structure the core comparison around four models and four prediction horizons resulting in 16 model-horizon comparisons. Note that ETTm1, Electricity and Weather are used as evaluation scenarios not the main comparison objects. MAE, RMSE, MSE, MAPE, prediction length, dataset and sample split statistics are reported on each result. Thus, analysis is concerned with predictive precision, horizon sensitivity, cross-dataset consistency and reporting transparency.

## 4 Models for Comparison

We compare four representative models which come from different application paradigms of forecasting. DLinear is applied as a linear baseline and is a simple trend-oriented forecasting. LightGBM is used as a feature-based machine learning baseline with gradient boosting decision trees [1]. LSTM is the classical RNN paradigm, and serves as a standard deep sequence modeling baseline [5, 6]. PatchTST is a Transformer-based forecasting method that learns patch-level representation to capture long-range temporal structure [3]. In the PatchTST implementation, reversible instance normalization is also relevant for handling distribution shift in time series inputs [4].

These four models together allow a fair comparison among the fields of linear forecasting, feature-based machine learning, recurrent neural networks, and transformer-based forecasting. It is this design that enables the benchmark to span multiple models, rather than narrow definitions of modeling paradigms.

## 5 Datasets and Forecasting Scope

We conduct experiments on three public benchmark datasets, namely ETTm1, Electricity and Weather for all models. These datasets are publicly available and were commonly used for long-term forecasting challenges [8, 9]. On each dataset, we evaluate every model on the same four prediction horizons ( $H = 96, 192, 336, 720$ ). Also we keep the raw dataset size and sample counts for training, validation, and test splits over time which are ordered chronologically apart from error metrics. It connects the forecasting performance with data scale and, further, makes the benchmark more outright.

## 6 Experimental Setup

We use a single experimental protocol to evaluate all models. Since this is a sequence generating prediction task, we make use of train/validation/test splits in chronological order to avoid information leakage from future observations into training. In all experimental settings, we use the same

dataset and prediction horizon for all models. The implementation is adapted and reproduced from the public Time-Series-Library framework [8], with model-specific adjustments when necessary.

The outputs of the experiments contain the details of MAE, RMSE, MSE, MAPE along with dataset, prediction length, rows and train samples and validation samples and test samples. In the case of LightGBM, the algorithm converts sequential observation into supervised learning samples by using lagged inputs so that the feature-based prediction model can be compared with neural sequence models under the same forecasting task. The benchmark of interest is forecasting errors and how this varies across models, horizons and datasets.

## 7 Evaluation Metrics and Targets of Testing

We employ four forecasting metrics: MAE, RMSE, MSE and MAPE. MAE, or Mean Absolute Error, measures the average absolute forecasting error. MSE imposes more serious penalties for larger deviations. The RMSE preserves the sensitivity of MSE, but remains on the original scale of the target variable. Mean Absolute Percentage Error: MAPE is a percentage based on absolute size, however it can be very unstable if target values are close to 0. Thus, the discussion focuses on MAE, RMSE, and MSE, while MAPE is used as an additional relative-error indicator.

Through the benchmark architecture, four targets are evaluated. In a sense, the predictive performance is assessed by measuring model errors in the same dataset and horizon. We study horizon sensitivity by checking how error varies across ( $H = 96, 192, 336, 720$ ). The cross-dataset consistency evaluates how those model-level patterns, which are visible in the training dataset are still interpretable on datasets such as ETTm1, Electricity and Weather. Dataset-scale transparency is supported by reporting rows and sample split statistics together with forecasting errors.

## 8 Results

The full numerical results for all combinations of model, dataset and horizon are reported in Online Appendix Table 1. We restrict ourselves to the main empirical patterns across the forecasting paradigms, prediction horizons, and benchmark datasets in order to keep the text concise. We perform the comparison based on two coarse-levels in four representative models, DLinear, LightGBM, LSTM and PatchTST with ( $H = 96, 192, 336, 720$ ) as prediction horizon on ETTm1, Weather and Electricity.

### 8.1 Overall Comparison of the Four Forecasting Paradigms

The complete metric results for all models, datasets and prediction horizons can be found in Appendix Table 1. The four forecasting paradigms on the other hand were clearly distinguishable in terms of numerical performance and model behaviour based on these results. In general, LightGBM performs best numerically across almost all settings, PatchTST is the strongest deep learning model, LSTM is simply a classic recurrent neural network baseline and DLinear implements a simple trend targeting reference.

For the benchmark, LightGBM leads all other methods. On ETTm1, with the growth of prediction horizon from 96 to 720, the MSE only increases from 0.424 to 0.6031. Weather: 0.7403 to 1.8591, Electricity: 0.4438 to 0.3474. These results show that, under the current benchmark setting, the feature-based tree model serves as a surprisingly strong non-neural baseline.

PatchTST, among deep learning models, offers the best overall performance. The errors grow with the prediction horizon, but it is a regular and interpretable pattern. An example during ETTm1, PatchTST gets a MSE of 3.15 and rises to 9.19; on Weather it rises from 186.95 to 304.23; on Electricity from 66,232 to 104,421. While the error grows with longer horizons, PatchTST keeps up its long-range structural modeling power better than the simpler deep baselines. This behavior is summarized in Figure 1 which displays test MSE of PatchTST across datasets and prediction horizons.

This heatmap summarizes PatchTST’s forecasting error on ETTm1, Weather and Electricity across four prediction horizons ( $H = 96, 192, 336, 720$ ) for each dataset. This shows how generally, with

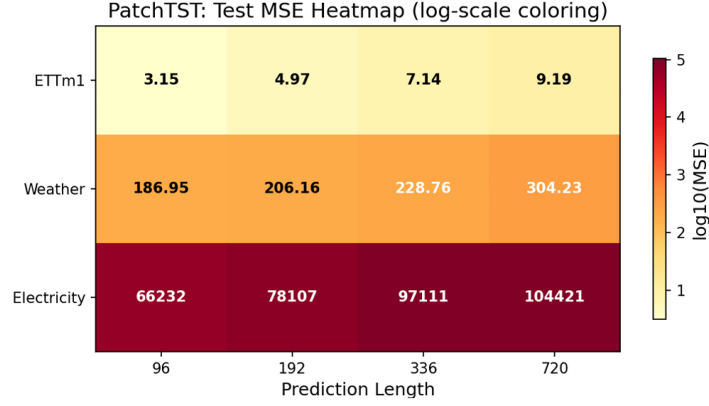


Figure 1: Test MSE heatmap of PatchTST across datasets and prediction horizons.

longer prediction horizons forecasting error is higher but the magnitude of the error varies across datasets.

From the classical recurrent neural network baseline we use LSTM. Its overall performance is more fragile compared to PatchTST due in particular longer horizon and more complicated forecasting environments. On ETTm1, its MSE stays at 11 to 14; on Weather, it is entirely approx. 982 to 992; and on Electricity, its MSE reaches approximately 9.34 million to 9.44 million. These results further indicate that the basic recurrent architecture is not competitive with simple patch-based Transformer modeling long-term forecast in the current benchmark.

DLinear behaves like a naive linear forecasting paradigm. It is best able to capture broad temporal patterns, but the error generally increases with prediction horizon. For example, its MSE on ETTm1 increases from 4.3678 to 8.3046, and RMSE increases from 2.0899 to 2.8818. On Weather, its MAE increases from 44.64 to 51.01. These results imply that DLinear can extract global trend, which is a good thing but less competitive than LightGBM and PatchTST in esoteric long-term forecasting setting.

Overall, taking a numerical look at the results, we end up with the following pattern: LightGBM has best overall metric performances, PatchTST is the strongest deep learning model. LSTM provides a big picture recurrent baseline and finally DLinear gives us simple linear reference model with obvious trend-following behaviour.

## 8.2 Horizon-wise Performance Trends

The finding also demonstrate that prediction horizon is a significant factor affecting the performance of the model. The plots show the average performance of all models on each prediction length  $H$  together with the training and evaluation datasets, per model family; as expected, most models start to perform worse when  $H$  increases although this degradation takes a slightly different form for each family of models.

The horizon effect is most obvious for PatchTST. All three datasets likewise show MSE, MAE, and RMSE steadily rising from  $H = 96$  to  $H = 720$ . Specifically, the RMSE increases from 1.77 to 3.03 on ETTm1, from 13.67 to 17.44 on Weather and from 257.36 to 323.14 on Electricity. That is, a Transformer-based architecture becomes exponentially more difficult with long forecasting windows.

For LSTM, the impact of prediction length is more data dependent. For ETTm1, the error does not increase strictly monotonically but enters a relatively high range compared to PatchTST. For Weather the relative error is steady, with MAPE around 6%. RMSE for Electricity increases from 3055.95 to 3072.79 as the horizon grows, showing the limitation of recurrent structures in more complex long-range forecasting tasks. LSTM is horizon-sensitive and therefore MAE and RMSE are plotted with respect to the four prediction lengths in Figure 2.

LSTM: MAE and RMSE vs. Prediction Horizon

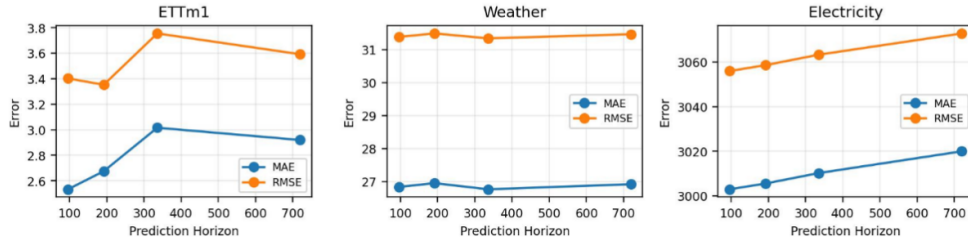


Figure 2: MAE and RMSE trends of LSTM across prediction horizons.

Variations of MAE and RMSE of LSTM (for ETTm1, Weather and Electricity datasets) when the prediction horizon augments from 96 to 720. It demonstrates that horizon-sensitivity of LSTM is dataset-dependent, much more stable on Weather and bigger errors on Electricity.

Specifically, for DLinear, a larger horizon results in poorer local fitting. This helps the model to preserve the general trend but results in a smoother prediction that is less sensitive to smaller peaks and troughs. In numerical, its MAE increases from 1.5018 to 2.1759 and RMSE increases from 2.0899 to 2.8818 on ETTm1 as well. This puts weight behind the interpretation that, while DLinear is supporting local trend modelling it tends to provide only a crude approximation when reconstructing long horizons with high detail.

With LightGBM, the horizon pattern is richer. The error generally increases with the length of prediction on both ETTm1 and Weather. On the other hand, Electricity RMSE decreases from 0.6662 to 0.5894 and MSE from 0.4438 to 0.3474 while MAPE increases here too. This indicates that the tree-based multi-output framework does not react to longer horizons in the same way these neural sequence models do.

All 16 model-horizon comparisons, taken together, illustrate that long horizons exacerbate forecasting difficulties but the degree and pattern of degradation differ by forecasting paradigm. Under the current context, we see that PatchTST has the most consistent horizon-wise error growth and clearly noticeable limitations at longer horizons in DLinear and LSTM, whereas LightGBM yields the most stable numerical performance.

### 8.3 Predictive Curves for Model Behavior

Prediction curves clearly differ in the four model families. DLinear exhibits a clear smoothing-dominant feature. It can track the overall trend of the series (particularly at short horizons), but slowly loses its capacity to model local movements as the horizon increases. This trend-following but smoothing behaviour at longest horizon is visible in Figure 3.

In the three subplots, we compare the original target series (blue) and forecasted one (orange). DLinear forces smoothing of the predictions, and therefore smoothens temporal peaks and troughs in the data sequence, despite being well capable of capturing the global movement (most significant on Electricity dataset).

On more regular series LSTM displays good sequence tracking. This becomes much worse in additional complex environments over longer horizons. The visual results are consistent with numerical ones: LSTM captures some of the most important trends, misses sharp spikes and large scale variability as in the Electricity dataset.

PatchTST has the strongest qualitative ability to conserve trend and periodic structure. The prediction curves are consistent with our assumption that patch-level representation learning benefits predictions by helping it recover clearer wider temporal patterns than simpler neural baselines. But its training curves are also signs of early overfitting, with the best validation epochs typically falling between 3 and 15. Moreover, since its parameter count increases from 1.77M to 5.29M, the benefit of long-range modelling comes at the expense of higher model complexity too.

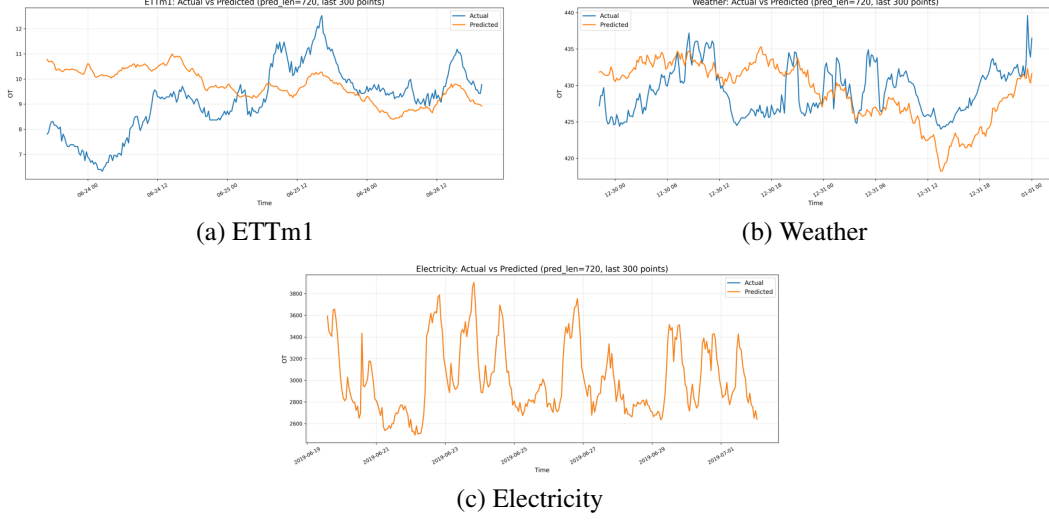


Figure 3: DLinear prediction versus ground truth at  $H = 720$  on ETTm1, Weather, and Electricity.

On LightGBM’s side, its metric table drives a strong and stable numerical performance. The representative LightGBM forecasts presented in Figure 4 (for the longest prediction horizon) support our numerical conclusion that even under this unified benchmark setting, tree models based on features are still able to compete with time series aspects.

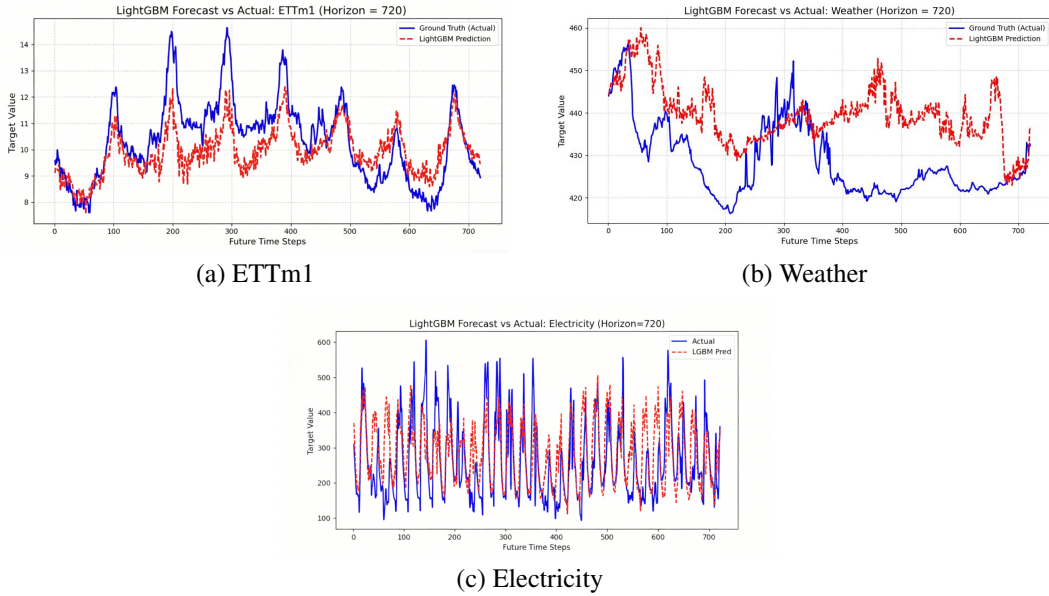


Figure 4: LightGBM forecast versus ground truth at  $H = 720$  on ETTm1, Weather, and Electricity.

The figure shows LightGBM’s predictions under the longest forecasting horizon,  $H = 720$ . LightGBM demonstrates strong numbers and compelling predictions relative to more easily established sequence baselines, suggesting that feature-based tree models are still far from dead even for long-run forecasting.

Figures 1–4 complement the numeric results with visual confirmation. PatchTST error patterns across multiple horizons, horizon sensitivity of LSTM, DLinear smoothing with long-horizon forecasting tendency and LightGBM strong fitting ability pattern. These figures reinforce the main conclusion that although all forecasting paradigms are described in columnar form for long-term prediction, they have different emphases.

All in all, the visual and numerical results confirm that the four models have distinct forecasting perspectives: DLinear emphasizes on trend extractions; LSTM on recurrent sequence memory; LightGBM on feature-driven regression; PatchTST long-range structural representation.

## 9 Discussion

In all cases, the largest source of difference among models is not whether or not they have lower or higher error but rather different assumptions about how time series should be modeled. DLinear assumes an almost trivial linear mapping can encapsulate much of the useful temporal structure, which is true to some extent in a global trend extraction sense (e.g. reduce an entire weekly cycle to 4 parameters), however this assumption begins breaking down rapidly when local fluctuations matter. LightGBM using a supervised learning over the features that can show timeseries forecasting concerning tree-based machine learning staying quite powerful [1, 2]. While LSTM models time dependence through recurrent hidden states and gated memory, the traditional recurrent structure is less competitive in long-horizon settings [5, 6]. PatchTST is based on patch-level Transformer representation learning and itself achieves the best deep learning performance—especially preserving long-range structure [3].

One of the major implications is that long-term time series forecast should not be assessed within a single model class. The current results show that a non-neural model (e.g. LightGBM) can surpass deep learning models in numerical metrics, while PatchTST remains the best neural approach overall. This means that every new forecasting benchmark for the future ought to compare predictive performance of both classical machine learning models and more modern deep architectures, rather than simply assuming deeper models come out on top.

Our results across horizons also demonstrate that the evaluation of model robustness must be conducted over multiple prediction lengths. What works for  $H = 96$ , may not work once we move to  $H = 720$ . In this study, PatchTST is seen to have an expected and interpretable increase in error with horizon length increasing, while LightGBM has a more dataset dependent response to the horizon. Therefore, reporting results at different horizons rather than a single prediction length is imperative in light of this.

The three benchmark datasets serve as evaluation scenarios instead of being the main objects of comparison. The four forecasting paradigms can be investigated under different temporal conditions, provided by ETTm1, Electricity, and Weather. While the exact numerical scale of errors differs from dataset to dataset, the overall model-layer trend remains informative: LightGBM is the strongest performing non-deep learning baseline, PatchTST is best among deep models, LSTM provides a recurrent baseline, and DLinear serves its purpose as a simple linear trend baseline.

## 10 Conclusion

In this paper, we give a comprehensive comparison of four representative models across various forecasting paradigms: DLinear, LightGBM, LSTM and PatchTST under a same long-term forecasting benchmark. The 4 prediction horizons, ( $H = 96, 192, 336, 720$ ), are denoted separately which leads to 16 core model-horizon comparisons. To better evaluate model performance under different data conditions, we also use three benchmark scenarios called ETTm1, Electricity and Weather.

The results indicate substantial paradigm-level differences. The DLinear has strong performance for global trend modeling but limited power in fine-grained fitting. LightGBM shows that model-based machine learning remains very competitive, with very strong and stable numerical performance. Although LSTM serves as a meaningful recurrent neural network baseline it is weaker than stronger modern architectures when predicting long-range time series data. PatchTST is the best deep learning model in the benchmark, yet it gains its competitive performance at the cost of higher model complexity and features a more evident ability of long-range structure modeling.

In total, and as this benchmark shows, their value resides in evaluating representative models from different forecasting paradigms. It is more useful to consider gains across families of architecture—linear models, feature-based machine learning, recurrent networks, and Transformer-based models—at different timescales for a much clearer picture of long time series forecasting performance.

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## A Full Numerical Results

Table 1 reports the full numerical forecasting results of DLinear, LightGBM, LSTM, and PatchTST across the three benchmark datasets and four prediction horizons. To keep the main text readable, only the main findings are discussed in Section 8, while the full model-dataset-horizon results are provided here.



Table 1: Full numerical results for all model-dataset-horizon combinations.

| Model    | Dataset     | H   | MSE      | MAE     | RMSE     |
|----------|-------------|-----|----------|---------|----------|
| DLinear  | ETTm1       | 96  | 4.3678   | 1.5018  | 2.0899   |
| DLinear  | ETTm1       | 192 | 6.1771   | 1.8267  | 2.4854   |
| DLinear  | ETTm1       | 336 | 7.5030   | 2.0408  | 2.7392   |
| DLinear  | ETTm1       | 720 | 8.3046   | 2.1759  | 2.8818   |
| DLinear  | Weather     | 96  | 79797.50 | 44.4641 | 282.4845 |
| DLinear  | Weather     | 192 | 85832.93 | 47.4646 | 292.9726 |
| DLinear  | Weather     | 336 | 87473.71 | 48.9980 | 295.7596 |
| DLinear  | Weather     | 720 | 86699.33 | 51.0102 | 294.4475 |
| DLinear  | Electricity | 96  | 25.8456  | 25.8456 | 5.0839   |
| DLinear  | Electricity | 192 | 26.5268  | 26.5268 | 5.1504   |
| DLinear  | Electricity | 336 | 26.1982  | 26.1982 | 5.1184   |
| DLinear  | Electricity | 720 | 26.3735  | 26.3735 | 5.1355   |
| LightGBM | ETTm1       | 96  | 0.4240   | 0.4725  | 0.6512   |
| LightGBM | ETTm1       | 192 | 0.4853   | 0.5020  | 0.6966   |
| LightGBM | ETTm1       | 336 | 0.5575   | 0.5338  | 0.7467   |
| LightGBM | ETTm1       | 720 | 0.6031   | 0.5652  | 0.7766   |
| LightGBM | Weather     | 96  | 0.7403   | 0.5693  | 0.8604   |
| LightGBM | Weather     | 192 | 0.9192   | 0.6611  | 0.9587   |
| LightGBM | Weather     | 336 | 1.2956   | 0.8142  | 1.1383   |
| LightGBM | Weather     | 720 | 1.8591   | 0.9696  | 1.3635   |
| LightGBM | Electricity | 96  | 0.4438   | 0.4092  | 0.6662   |
| LightGBM | Electricity | 192 | 0.4086   | 0.3949  | 0.6392   |
| LightGBM | Electricity | 336 | 0.3610   | 0.3820  | 0.6009   |
| LightGBM | Electricity | 720 | 0.3474   | 0.3903  | 0.5894   |
| LSTM     | ETTm1       | 96  | 11.5765  | 2.5335  | 3.4024   |
| LSTM     | ETTm1       | 192 | 11.2426  | 2.6751  | 3.3530   |
| LSTM     | ETTm1       | 336 | 14.1010  | 3.0161  | 3.7551   |
| LSTM     | ETTm1       | 720 | 12.9016  | 2.9197  | 3.5919   |
| LSTM     | Weather     | 96  | 985.0976 | 26.8366 | 31.3863  |
| LSTM     | Weather     | 192 | 991.7117 | 26.9508 | 31.4915  |
| LSTM     | Weather     | 336 | 982.3721 | 26.7654 | 31.3428  |
| LSTM     | Weather     | 720 | 990.2131 | 26.9188 | 31.4677  |
| LSTM     | Electricity | 96  | 9338827  | 3002.97 | 3055.95  |
| LSTM     | Electricity | 192 | 9354807  | 3005.54 | 3058.56  |
| LSTM     | Electricity | 336 | 9383646  | 3010.25 | 3063.27  |
| LSTM     | Electricity | 720 | 9442008  | 3019.97 | 3072.79  |
| PatchTST | ETTm1       | 96  | 3.15     | 1.30    | 1.77     |
| PatchTST | ETTm1       | 192 | 4.97     | 1.67    | 2.23     |
| PatchTST | ETTm1       | 336 | 7.14     | 2.03    | 2.67     |
| PatchTST | ETTm1       | 720 | 9.19     | 2.32    | 3.03     |
| PatchTST | Weather     | 96  | 186.95   | 10.00   | 13.67    |
| PatchTST | Weather     | 192 | 206.16   | 10.54   | 14.36    |
| PatchTST | Weather     | 336 | 228.76   | 11.10   | 15.12    |
| PatchTST | Weather     | 720 | 304.23   | 12.83   | 17.44    |
| PatchTST | Electricity | 96  | 66232    | 183.14  | 257.36   |
| PatchTST | Electricity | 192 | 78107    | 198.05  | 279.48   |
| PatchTST | Electricity | 336 | 97111    | 220.81  | 311.63   |
| PatchTST | Electricity | 720 | 104421   | 240.87  | 323.14   |

**Our Presentation Video Link :** <https://youtu.be/tTOUs6bftE4>